Ideation:

- **API Breaking-Change Impact Simulator**
    - Compare two versions of an API specification to detect breaking changes, such as removed endpoints, renamed fields, or changes to required parameters
    - Explain which changes could affect existing applications and provide examples of potentially affected requests or responses
- **LLM Prompt-Injection Attack Replay Lab**
    - Test malicious prompts against different LLM configurations to evaluate whether they can override instructions or cause unintended behavior
    - Compare baseline and defended configurations to identify vulnerabilities and measure how well different defenses work
- **Duplicate Customer Record Merge Recommendation Engine**
    - Identify potentially duplicated customer records using fuzzy matching across fields such as names, email addresses, and phone numbers
    - Recommend which records may be duplicates and explain the matching evidence so users can review suggestions before merging records
- **Subscription Cancellation Reason Pattern Miner (Undone)**
    - Analyze customer cancellation feedback using natural language processing to group similar responses into themes, such as pricing, missing features, or technical issues
    - Visualize common cancellation reasons and emerging trends, with comparisons across subscription plans or customer segments

After considering these ideas, we chose Undone, a Subscription Cancellation Feedback Analyzer. We wanted to explore a project that uses AI and data analysis to find useful patterns in text rather than simply generating content from a prompt. The application will help users understand common reasons customers cancel subscriptions and identify trends that could inform future product improvements.

Idea Description:

Undone: An AI-Powered Subscription Cancellation Feedback Analyzer

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Abstract

Team Undone - Team 4

Undone is a web application that helps businesses understand why customers cancel their subscriptions by analyzing cancellation feedback and identifying common patterns. Users will be able to upload cancellation data, which the application will process to group similar responses into categories such as pricing concerns, missing features, and technical issues. The application will also display visualizations of the most common cancellation reasons, show how these reasons change over time, and allow users to compare feedback across different subscription plans or customer groups. By organizing individual responses into more understandable patterns, Undone aims to help businesses identify areas that may need improvement instead of having to manually review large amounts of customer feedback.

Introduction

Subscription-based services often receive feedback from customers who decide to cancel their subscriptions. This feedback can contain useful information about problems with pricing, product features, customer experience, or technical issues. However, when a business receives a large number of responses, it can be difficult to read through all of them and identify the most common concerns. Looking at individual responses may also make it harder to recognize patterns that appear across different customers or change over time.

Undone is designed to address this problem by using text analysis and data visualization to organize cancellation feedback. Instead of requiring users to manually sort through every response, the application will group similar feedback and present the results through a dashboard. This will allow users to understand the main reasons customers report for cancelling and explore the feedback in more detail.

Motivation / Why This Idea

The problem we are trying to solve is that businesses may collect a lot of cancellation feedback but not always have an efficient way to analyze it. Reading each response individually can take

time, especially when responses are written in different ways even though they describe similar problems. For example, one customer might say that a subscription costs too much, while another might say that the monthly price is not worth the features provided. These responses are worded differently but may represent a similar concern.

We want to use text analysis and machine learning to identify these similarities and organize responses into groups. This is interesting because it allows us to apply AI to a practical problem while also using data analysis to make the results easier to understand. We also want to examine how cancellation reasons vary over time and between customer groups. For example, businesses could investigate whether pricing complaints are becoming more common or whether customers on a particular subscription plan report different problems.

Another goal is to make the results useful rather than only showing the output of a machine learning model. By providing charts, example responses, and possible actions to investigate, Undone will help users understand what the data suggests. These patterns will not necessarily prove why customers cancelled, but they can help businesses decide which problems may deserve further attention.

Proposed Features

- **Cancellation Feedback Upload**
    - Allow users to upload a CSV file containing cancellation feedback and related information, such as cancellation dates and subscription plans
    - Process and organize the uploaded data for analysis
- **Feedback Pattern Detection**
    - Use text processing and machine learning to group similar cancellation responses
    - Identify common themes such as pricing, missing features, technical problems, and customer service
- **Cancellation Analytics Dashboard**
    - Display the most common cancellation reasons and the percentage of responses associated with each category
    - Use charts to make the results easier to understand
- **Trend Analysis**
    - Show how cancellation reasons change over time
    - Help users identify whether particular concerns are becoming more common
- **Customer Segment Comparison**
    - Allow users to compare cancellation patterns across subscription plans or other available customer groups
    - Show how different groups may report different concerns
- **Evidence-Based Recommendations**
    - Suggest possible areas for improvement based on the patterns found in the data

- Include supporting statistics and example responses so users can understand why a recommendation was made

Application Functionality

The first step will be for a user to upload a CSV file containing cancellation feedback. The application will process the file and prepare the responses for analysis. It will clean and organize the text before using a text analysis model to identify responses that discuss similar topics. These responses will then be grouped into categories that represent common cancellation reasons.

After the analysis is complete, users will be able to view a dashboard showing the most frequently identified reasons for cancellation. The dashboard will include charts and percentages to summarize the results. Users will also be able to explore individual responses within each group to understand the feedback behind the identified patterns.

The application will also use the dates and other information included in the uploaded data to compare cancellation reasons over time and across subscription plans. For example, users could examine whether pricing-related feedback has increased in recent months or whether customers on different plans report different issues. This will allow users to explore the data beyond the overall summary.

Finally, Undone will provide possible recommendations based on the analysis. For example, if many responses mention missing features, the application might suggest investigating those feature requests. Recommendations will be connected to the underlying feedback and statistics so users can see what led to each suggestion. The application will present these results as patterns and possible areas for improvement rather than assuming that feedback alone proves the cause of a cancellation.

References

No external datasets or research papers have been selected yet.